

Worksheet

Let X be a Δ -complex. Including each simplex of X as a singular simplex gives a chain map

$$\iota : C_{\bullet}^{\Delta}(X) \hookrightarrow C_{\bullet}(X),$$

and hence a comparison map $\iota_* : H_k^{\Delta}(X) \rightarrow H_k(X)$. We are proving that ι_* is an isomorphism, by induction on skeleta.

Q1. Base case. Show that $H_k^{\Delta}(X^0) \cong H_k(X^0)$.

Hint: compute $H_k^{\Delta}(X^0)$ first.

Q2. Set up the induction.

(a) Write down maps comparing the long exact sequence of the pair (X^n, X^{n-1}) in Δ -homology to the same long exact sequence in singular homology.

(b) Write down a fact about relative homology which would allow us to finish the induction.

(c) Compute $H_k^\Delta(X^n, X^{n-1})$.

Q3. What's left is computing $H_k(X^n, X^{n-1})$. We build a good open cover to help. Let $\{\sigma_i\}_{i \in I}$ be the set of n -simplices of X , U_i a neighbourhood of σ_i , and V a neighbourhood of X^{n-1} .

Properties of the open cover $\mathcal{U} = \{V\} \cup \{U_i\}_{i \in I}$

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(a) Prove that

$$H_k\left(\frac{C_{\bullet}^{\mathcal{U}}(X^n)}{C_{\bullet}(V)}\right) \cong H_k(X^n, X^{n-1}).$$

What properties do we need $\mathcal{U}$ to have to make this work?

(b) Find the kernel of the surjective map

$$\bigoplus_{i \in I} C_{\bullet}(U_i) \longrightarrow \frac{C_{\bullet}^{\mathcal{U}}(X^n)}{C_{\bullet}(V)}.$$

What does this tell us about $H_k(X^n, X^{n-1})$?

Q4. Compute $H_k(U_i, U_i \cap V)$.

Hint 1: what are U_i and $U_i \cap V$ as topological spaces?

Hint 2: go back and add an extra property to your definition of the open cover $\mathcal{U}$.